



Subclasses and Inheritance

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Lecture #	2
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Week #	3

A typical Java class

```
public class Employee {
    private String name;
    private double salary;

    /* Constructors ... */

    // Mutator methods
    public boolean setName(String s) { ... }
    public boolean setSalary(double x) { ... }

    // Accessor methods
    public String getName() { ... }
    public String getSalary() { ... }

    // Other methods
    public double bonus(float percent) {
        return (percent/100.0) * salary;
    }
}
```

What do we have here

- An `Employee` class
- Two private instance variables
- Some constructors to set up the object
- Accessor and Mutator methods to set instance variables
- A public method to compute bonus

Subclasses

- Managers are special types of employees with extra features

```
public class Manager extends Employee {
    private String secretary;
    public boolean setSecretary(String s) { ... }
    public String getSecretary() { ... }
}
```

- `Manager` object inherit other fields and methods from `Employee`
 - Every `Manager` has a `name`, `salary` and methods to access and manipulate these
- `Manager` is a *subclass* of `Employee`
- `Manager` objects do not automatically have access to private data of the parent class
 - Common to extend a parent class written by someone else
- How can a constructor for `Manager` set instance variables that are private to `Employee`?
 - Use parent class's constructor using `super`

```
public class Employee {
    ...
    public Employee(String n, double s) {
        name = n;
        salary = s;
    }
    public Employee(String n) {
        this(n, 500.00);
    }
}

public class Manager extends Employee {
    ...
    public Manager(String n, double s, String sn) {
        super(n, s);
        secretary = sn;
    }
}
```

Inheritance

- In general, subclass has more features than parent class
 - Subclass inherits instance variables, methods from the parent class
- Every `Manager` is an `Employee`, but every `Employee` is not a `Manager`
- Can use a subclass in place of a superclass

```
Employee e = new Manager(...);
```

- But the following will not work

```
Manager m = new Employee(...);
```

Summary

- A subclass extends a parent class
- Subclass inherits instance variables and methods from the parent class
- Subclass can add more instance variables and methods
 - Can also override methods — later
- Subclasses cannot see private components of parent class
- Use `super` to access the constructor of parent class